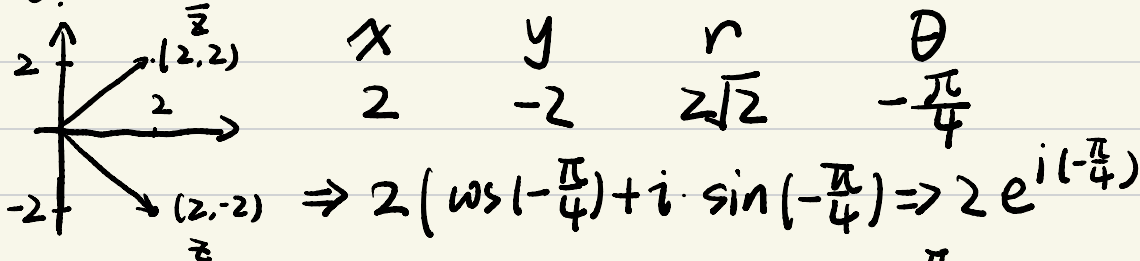


作业 1 重写 (Section 4 + Section 12)

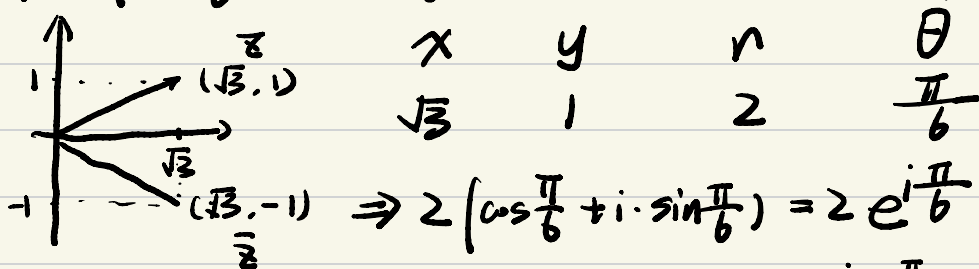
P51 Section 4

10. $2 - 2i \Rightarrow (2, -2)$



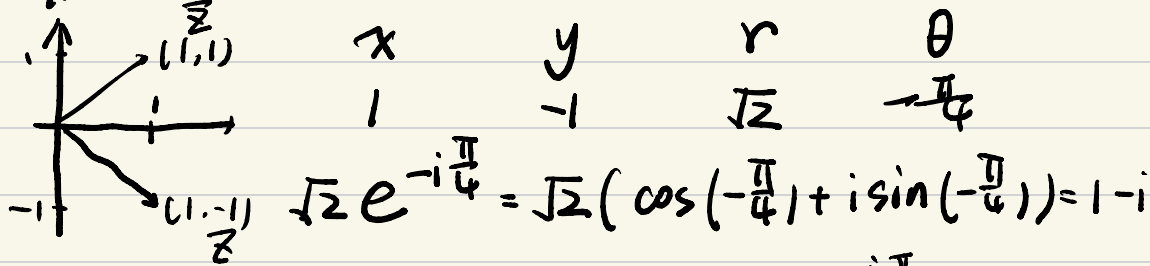
conjugate: $\bar{z} = 2 + 2i = 2e^{i\frac{\pi}{4}}$

11. $2(\cos\frac{\pi}{6} + i \sin\frac{\pi}{6}) \Rightarrow \sqrt{3} + i \Rightarrow (\sqrt{3}, 1)$



conjugate: $\bar{z} = \sqrt{3} - i = 2e^{i(-\frac{\pi}{6})}$

17. $\sqrt{2}e^{-i\frac{\pi}{4}} \Rightarrow (1, -1)$



conjugate = $1 + i = \sqrt{2}e^{i\frac{\pi}{4}}$

$$18. 3e^{i\frac{\pi}{2}} \Rightarrow (0, 3)$$

	x	y	r	θ
	0	3	3	$\frac{\pi}{2}$

$$-3i \bar{z}(0, -3) \Rightarrow 3e^{i\frac{\pi}{2}} = 3(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2}) = 3i$$

$$\text{conjugate: } \bar{z} = -3i = 3e^{-i\frac{\pi}{2}}$$

P69 Section 12

$$6. \cos(i \ln 5) = \frac{e^{-\ln 5} + e^{\ln 5}}{2} = \frac{5 + \frac{1}{5}}{2} = \frac{13}{5}$$

$$10. \sin(i \ln i) \Rightarrow \ln i = \ln e^{i\frac{\pi}{2}} = i \cdot \frac{\pi}{2}$$

$$= \sin(i \cdot i \frac{\pi}{2}) = \sin(-\frac{\pi}{2}) = -1$$

$$17. \int e^{ax} \cos bx \, dx = \frac{e^{ax}(a \cos bx + b \sin bx)}{a^2 + b^2}$$

$$\text{prove: } \int e^{(a+ib)x} \, dx = \int e^{ax} \cdot e^{ibx} \, dx$$

$$e^{ibx} = \cos(bx) + i \sin(bx)$$

$$\Rightarrow \int e^{(a+ib)x} dx = \int e^{ax} \cos(bx) dx + \int e^{ax} \cdot i \cdot \sin(bx) dx$$

$$\int e^{(a+ib)x} dx = \frac{e^{(a+ib)x}}{a+ib} = \frac{e^{(a+ib)x} (a-ib)}{a^2+b^2}$$

$$= \frac{e^{ax} (\cos bx + i \sin bx) (a-ib)}{a^2+b^2}$$

$$= \frac{e^{ax}}{a^2+b^2} (a \cos bx + i a \sin bx - i b \cos bx + b \sin bx)$$

$$= \frac{e^{ax}}{a^2+b^2} (a \cos bx + b \sin bx \xrightarrow{\text{real}} + i (a \sin bx - b \cos bx) \xrightarrow{\text{imaginary}})$$

for real:

$$\int e^{ax} \cos bx dx = \frac{e^{ax}}{a^2+b^2} (a \cos bx + b \sin bx)$$

经检验，等式成立

P71 Section 12

$$\begin{aligned} 33. \tan i &= \frac{\sin i}{\cos i} = \frac{e^{-1} - e}{(e^{-1} + e)i} \\ &= \frac{1 - e^2 i^2}{1 + e^2 i^2} \\ &= \frac{e^2 - 1}{e^2 + 1} i \end{aligned}$$

$$\begin{aligned} 34. \sin\left(i \frac{\pi}{2}\right) &= \frac{e^{-\frac{\pi}{2}} - e^{\frac{\pi}{2}}}{2i} \\ &= \frac{e^{-\frac{\pi}{2}} - e^{\frac{\pi}{2}}}{2i} \cdot \frac{i^2}{i^2} \\ &= \frac{e^{\frac{\pi}{2}} - e^{-\frac{\pi}{2}}}{2} i \end{aligned}$$